WORKED SOLUTION

STEP II 1996, Question 13

Probability and Statistics • Random Walks on a Grid

STEP II 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q13

The key idea

The identity is Vandermonde's: compare coefficients of t^n on the two sides. For the walk, the useful quantity is a single number that every step changes by exactly one: how many blocks north plus how many blocks west you are of my hotel. Each of my steps raises it by 1, and each of my friend's steps lowers it by 1. So we can only meet at one moment, along one diagonal line of intersections, and the meeting probability becomes exactly the sum in the identity.

Solution

The identity

The coefficient of t^r in $(1+t)^n$ is $\binom{n}{r}$. A term in t^n in the product $(1+t)^n(1+t)^n$ comes from choosing t^r from the first factor and t^{n-r} from the second, for some $r = 0, 1, \ldots, n$. So the coefficient of t^n on the left is

$$\sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r},$$

and on the right it is $\binom{2n}{n}$. Equating the two gives the required identity. The same argument applied to $(1+t)^a(1+t)^b = (1+t)^{a+b}$ gives the more general version

$$\sum_{r} \binom{a}{r}\binom{b}{n-r} = \binom{a+b}{n}, \qquad (*)$$

which we shall need later.

Where we can meet

Put my hotel at the origin, with x measured east and y north in blocks, so my friend's hotel is at $(-n, n)$. For any position define

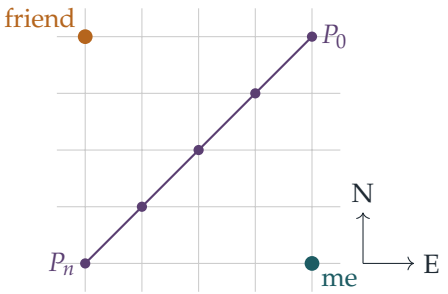
$$D = y - x,$$

the number of blocks north plus the number of blocks west of my hotel. Going north or going west raises D by 1; going south or going east lowers it by 1. So after walking for t minutes, I have $D = t$, and my friend, who starts at $D = 2n$, has $D = 2n - t$. This holds at every instant, mid-block as well, because D changes steadily as we walk.

To be at the same place at the same time we need equal values of D, so $t = 2n - t$, giving $t = n$. Hence we can meet only after exactly n minutes, and then at an intersection with $y - x = n$. My n steps keep me inside the region $-n \le x \le 0$, $0 \le y \le n$, so the possible meeting points are

$$P_r = (-r,\, n - r), \qquad r = 0, 1, \ldots, n,$$

which are the $n + 1$ intersections on the diagonal of the square joining the intersection n blocks north of my hotel to the one n blocks west of it. (In particular we can never pass each other in the middle of a block.)



The sketch shows $n = 4$: the meeting points are the dots on the diagonal.

The probability of meeting

I am at P_r after n minutes exactly when r of my n steps were west and $n - r$ were north. Each sequence of n steps has probability 2^{-n}, so

$$\mathrm{P}(\text{I am at } P_r) = \binom{n}{r}\frac{1}{2^n}.$$

My friend reaches $P_r = (-r, n - r)$ from $(-n, n)$ after n steps exactly when $n - r$ of them were east and r were south, which has probability $\binom{n}{n-r}2^{-n}$. Our walks are independent, and the events "we are both at P_r" for different r are mutually exclusive, so

$$\mathrm{P}(\text{meet}) = \sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r}\frac{1}{4^n} = \frac{1}{4^n}\binom{2n}{n},$$

by the identity.

Leaving $2k$ minutes late

Now measure time T from when my friend sets off. For $T \ge 2k$ I have walked for $T - 2k$ minutes, so I have $D = T - 2k$, while my friend has $D = 2n - T$. These are equal only

when $T = n + k$. Then I have taken $n - k$ steps (here $k \le n$, so this is a non-negative whole number) and my friend has taken $n + k$. Since $n + k \ge 2k$, my friend cannot reach my side of the diagonal while I am still asleep in my hotel. Again we can only meet at an intersection, never mid-block.

After $n - k$ steps, r of them west, I am at $(-r, n - k - r)$, with probability $\binom{n-k}{r}2^{-(n-k)}$. My friend is at the same point if $n - r$ of her $n + k$ steps were east and $k + r$ were south, which has probability $\binom{n+k}{n-r}2^{-(n+k)}$. Summing over r as before,

$$\mathrm{P}(\text{meet}) = \frac{1}{2^{n-k}\,2^{n+k}} \sum_{r=0}^{n-k} \binom{n-k}{r}\binom{n+k}{n-r} = \frac{1}{4^n}\binom{2n}{n},$$

by $(*)$ with $a = n - k$ and $b = n + k$.

? **Ask yourself** Why should oversleeping make no difference at all?

Look at the walk from my friend's point of view, measuring my position relative to hers. Each of her steps (south or east) and each of my steps (north or west) moves me, relative to her, one block north or one block west, with probability $\frac{1}{2}$ each, independently. We start n blocks apart in each direction, and we meet exactly when the total of $2n$ steps (mine plus hers) consists of n "north" and n "west" relative moves. It doesn't matter who makes which steps, so the answer is $\binom{2n}{n}/2^{2n}$ however the $2n$ steps are shared out, whether n and n, or $n - k$ and $n + k$. The condition on k just ensures that I really do get to walk my share. For large n the probability is about $1/\sqrt{\pi n}$, which is small, and it is the same whether or not I oversleep by an even number of minutes.

$n = 1$, one minute late

Now I have $D = T - 1$ and my friend has $D = 2 - T$, which are equal only at $T = \frac{3}{2}$, a time when we are both halfway along a block. So we can *never* be at the same intersection at the same time, and if meeting means meeting at an intersection, the probability is 0. The parity has changed: the total number of steps is now odd.

If we count passing each other in the middle of a block (as two real people would), we meet at $T = \frac{3}{2}$ when we walk along the same block in opposite directions. At $T = 1$ I am still at my hotel $(0, 0)$ and my friend is at $(0, 1)$ or $(-1, 0)$. We then meet if she is at $(0, 1)$ and goes south while I go north, or she is at $(-1, 0)$ and goes east while I go west. Each of these has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$, so

$$\mathrm{P}(\text{meet mid-block}) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Checks

- For $n = 1$ with no delay we meet if I go north and she goes east, or I go west and she goes south: probability $\frac{1}{4} + \frac{1}{4} = \frac{1}{2} = \binom{2}{1}/4$.
- By computer, we enumerated every pair of walks for $n \le 5$ and all allowed k, tracking both walkers every half minute. The meeting probabilities were $\frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \frac{35}{128}, \frac{63}{256}$

whatever the even delay, always at the single predicted time. With $n = 1$ and a one-minute delay, the only meetings were mid-block at $T = \frac{3}{2}$, with total probability $\frac{1}{4}$.

- For $n = 100$, $\binom{200}{100}/4^{100} \approx 0.0563$, close to $1/\sqrt{100\pi} \approx 0.0564$.

Common slips

- Assuming we could meet anywhere in the square at any time. The quantity $y - x$ pins the meeting down to one time and one diagonal.
- Squaring the probabilities without summing over the meeting points, or forgetting that my friend's step counts at P_r are swapped (she needs $n - r$ east steps, not r).

Beyond STEP

The relative-position argument is a coupling: two independent random walks are replaced by one walk of combined length, which is why $\binom{2n}{n}/4^n$ keeps reappearing. The same quantity is the probability that a simple symmetric random walk on a line is back at the origin after $2n$ steps, and its $1/\sqrt{\pi n}$ decay (from Stirling's formula) is the key to Pólya's theorem that such walks in one and two dimensions return to the start with probability 1. Grimmett and Welsh, *Probability: An Introduction*, treats random walks and these return probabilities.

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